SINGAPORE SCHOOL KINDERLAND

Computer Science — Comprehensive Master Study Guide

Cambridge IGCSE & O Level / Secondary 2 • Topic 1.2.3 & 1.2.4: Data Compression

1. Introduction to Data Compression (Section 1.2.3)

Sound and image files contain large amounts of digital binary data. Storing and transmitting uncompressed digital files consumes massive amounts of storage and transmission capacity. Therefore, files must be **compressed**.

> **Definition — Data Compression:**
> Data compression is the process of reducing the size of a computer file by using compression algorithms that either eliminate redundant data or permanently remove unnoticeable detail.

Why is File Compression Necessary? (Five Core Reasons)

1. **Saving Storage Space:** Smaller file sizes save valuable storage capacity on primary and secondary storage media such as Hard Disk Drives (HDDs), Solid-State Drives (SSDs), USB flash drives, and memory cards.
2. **Faster Streaming:** It reduces the time taken to stream audio and video files across the internet. Without compression, real-time video buffering and music streaming would stutter and freeze.
3. **Faster Uploads, Downloads & File Transfers:** Smaller files transfer across local networks and the internet significantly faster because fewer data packets have to be transmitted.
4. **Bandwidth Efficiency:**
   - **Network Bandwidth** is defined as the *maximum rate of transfer of data across a network, measured in bits per second (bps)*.
   - For example, downloading an uncompressed 100 KiB file requires 100 KiB/s of bandwidth, whereas a compressed 10 KiB file requires only 10 KiB/s. Compression enables more users to share the same network without slowdowns.
5. **Cost Reduction:**
   - Cloud storage service providers (e.g., Google Drive, AWS, OneDrive) charge fees directly based on the storage volume occupied.
   - Internet Service Providers (ISPs) often charge fees or apply fair-usage data caps based on the total quantity of data downloaded. Smaller files lower both storage and data costs.

2. Methods of File Compression: Lossy vs. Lossless (Section 1.2.4)

Compression techniques fall strictly into two fundamental categories: **Lossy Compression** and **Lossless Compression**.

Feature	Lossy Compression	Lossless Compression
How it Works	The algorithm permanently eliminates unnecessary or redundant data that human senses cannot easily detect.	The algorithm rearranges or encodes identical repeated data without losing any original binary information.
Restoration to Original	**NO.** The original uncompressed file cannot be reconstructed once compressed.	**YES.** The original uncompressed file can be completely reconstructed bit-for-bit.

Quality / Detail Loss	Some loss of detail occurs, but it is acceptable for general human perception.	Zero loss of quality or data. Perfect 100% reproduction.
File Size Reduction	**Very High** (up to 85% to 90% reduction in file size).	**Moderate** (typically 30% to 50% reduction in file size).
Typical File Formats	**MP3** (Audio), **MP4** (Video/Multimedia), **JPEG** (Images).	**PNG**, **GIF**, **ZIP**, **FLAC**, Run-Length Encoded files.
Suitable Applications	Photographs, music streaming, internet video streaming.	Spreadsheets, text documents, databases, computer executable programs (`.exe`), source code.

Exam Warning — Critical Concept:
Never use Lossy compression on program executable files (`.exe`), spreadsheets, financial records, or word processing documents! Discarding even a single binary bit will corrupt code instructions or alter financial formulas, causing program crashes or fatal calculation errors.

3. In-Depth Study of Lossy Compression Algorithms

Lossy compression algorithms achieve dramatic size reductions by taking advantage of the physical limitations of human biological perception (eyes and ears).

A. MP3 (MPEG-3) Audio Compression

- **Purpose:** Used for playing and streaming audio tracks on computers, tablets, and smartphones.
- **Compression Ratio:** Reduces the file size of a CD-quality audio track by **about 90%** (a 50 MB track becomes ~5 MB).
- **How it Works — Perceptual Music Shaping:**
  1. **Removal of Inaudible Frequencies:** Sounds outside the normal range of human hearing (below 20 Hz or above 20 kHz / 20,000 Hz) are permanently stripped away.
  2. **Auditory Masking (Perceptual Shaping):** If two sounds are played simultaneously, the human ear can only register the louder sound. The softer sound is completely masked, so the algorithm eliminates the softer sound without the listener noticing.

B. MP4 (MPEG-4) Multimedia Compression

- **Purpose:** Used for storing and streaming **multimedia files** (sound, music, video footage, photographs, and animations).
- **Advantage:** Retains high acceptable visual and audio quality while producing small file sizes suitable for high-definition internet movie streaming (e.g., YouTube, Netflix).

C. JPEG Image Compression

- **Purpose:** Standard compression algorithm for photographic bitmap images produced by digital cameras and smartphones.
- **Raw Camera Files:** Digital camera sensors produce huge, temporary *raw bitmap files*. Converting them to JPEG produces a compressed file and permanently discards the original raw data.
- **Two Core Concepts of JPEG Compression:**
  1. **Brightness vs. Colour Sensitivity:** The human eye contains far more rod cells (brightness detectors) than cone cells (colour detectors). Consequently, human eyes detect variations in image brightness (luminance) much better than subtle variations in colour shades (chrominance).
  2. **8 × 8 Pixel Block Averaging:** The image is divided into small 8 × 8 pixel blocks. The algorithm separates pixel colour from brightness, and discards subtle colour variations across adjacent pixels while preserving brightness levels. This achieves huge size reductions with no noticeable deterioration to human viewers.

4. Lossless Compression & Run-Length Encoding (RLE)

When files cannot afford any data loss, **Run-Length Encoding (RLE)** is the primary lossless compression technique used.

> **Definition — Run-Length Encoding (RLE):**
> RLE is a form of reversible, lossless data compression that reduces the size of a repeating string of identical adjacent data units (such as repeated characters in text or repeated pixel colours in an image).

How RLE Encodes Data (Two-Value Encoding)

A repeating sequence (a "run") is encoded into a pair of values:

- **Value 1 (Run Count):** Represents the number of consecutive identical items.
- **Value 2 (Data Value / Code):** Represents the actual data item or character (such as its ASCII code or colour value).

Worked Example 1: RLE on Simple Text Data

Consider the text string: `aaaaabbbbccddddd`

• **Original Size:** 5 'a' + 4 'b' + 2 'c' + 5 'd' = 16 characters = **16 Bytes** (assuming 1 Byte/character in ASCII).

• **RLE Encoding:**

- 5 'a' (ASCII 97) → `05 97` (2 bytes)
- 4 'b' (ASCII 98) → `04 98` (2 bytes)
- 2 'c' (ASCII 99) → `02 99` (2 bytes)
- 5 'd' (ASCII 100) → `05 100` (2 bytes)

• **Compressed Size:** 4 pairs × 2 bytes = **8 Bytes**.

• **Result:** File size is reduced by **50%** without losing a single character!

Handling Non-Repeating Data using a "Flag" Byte

When data does not repeat frequently (e.g., `abcdefgh`), standard RLE would double the file size by writing `01 'a' 01 'b'...`. To prevent this, algorithms use a **Flag byte** (e.g., `255`):

- A **Flag byte (255)** indicates that a run follows. The next byte is the *repetition count*, followed by the *character code*.
- When there is no flag, bytes are read at their face value as non-repeating single characters.

> **Worked Example 2: RLE with Flag Byte (255)**
> Consider string: `a a a a a b c d e f g h i j k l`
> • **Original String:** 16 characters = **16 Bytes**.
> • **Using Flag Byte 255:**
> `255 05 97 98 99 100 101 102 103 104 105 106 107 108`
> (Here, `255 05 97` encodes the five 'a's, and the remaining 11 single characters are stored directly).
> • **Compressed Size:** 3 bytes (flag pair) + 11 bytes = **14 Bytes**.
> • **Conclusion:** Using flags prevents RLE from inflating the size of mixed or non-repeating data.

RLE with Images (Black & White and Coloured)

Images contain large contiguous blocks of identical colours (e.g., backgrounds, borders), making them ideal for RLE.

> **Worked Example 3: RLE on a Black & White Bitmap Image (Letter 'F')**
> Suppose an 8 × 8 pixel grid (64 pixels total) displays the letter 'F':
> • **Uncompressed File Size:** 8 × 8 = 64 pixels. At 1 Byte per pixel = **64 Bytes**.
> • **Row-by-Row RLE Encoding (W = White pixel, B = Black pixel):**
> Row 1 (8 Black): `8B`
> Row 2 (2 Black, 6 White): `2B 6W`
> Row 3 (2 Black, 6 White): `2B 6W`
> Row 4 (6 Black, 2 White): `6B 2W`
> Row 5 (2 Black, 6 White): `2B 6W`
> Row 6 (2 Black, 6 White): `2B 6W`
> Row 7 (8 White): `8W`
> Row 8 (8 White): `8W`
> • **RLE Data Stream:** `8B 2B 6W 2B 6W 6B 2W 2B 6W 2B 6W 8W 8W` (total 13 pairs = 26 bytes).
> • **Size Reduction:** Reduced from 64 Bytes down to 26 Bytes (over 59% reduction).

5. Extension: Binary Coded Decimal (BCD)

> **What is Binary Coded Decimal (BCD)?**
> BCD is a number encoding system where each decimal digit from 0 to 9 is represented by its own fixed **4-bit binary nibble** (from 0000 to 1001).

- **Primary Applications:** Electronic handheld calculators, digital clocks, micro-oven displays, and financial applications where exact decimal digit representation is required without binary fractional rounding errors.

- **Worked Example: Representing 180.3 in BCD:**

Decimal Digit	1	8	0	.	3
4-Bit BCD Nibble	`0001`	`1000`	`0000`	[Decimal Point]	`0011`

→ Final BCD String: `0001 1000 0000 . 0011`

6. Key Terms & Definitions Summary Checklist

Term	Exam Definition & Meaning
Data Compression	Reducing the size of a file to save storage space and decrease transmission time across a network.
Bandwidth	The maximum rate of transfer of data across a network, measured in bits per second (bps).
Lossy Compression	Compression technique that permanently removes unnecessary data; the original file cannot be reconstructed.
Lossless Compression	Compression technique where no data is lost; the original file can be reconstructed with 100% precision.
Perceptual Music Shaping	Lossy audio technique that eliminates inaudible sound frequencies and softer sounds masked by louder sounds.
Run-Length Encoding (RLE)	Lossless compression method that replaces adjacent repeating identical items with a single run count and data code.